ModelIt!

Student Activity Pack

Where Does All the Plastic Go?

Modeling Plastic Pollution Pathways, Microplastic Accumulation, and Bioaccumulation

Grade 12L2 | NGSS HS-ESS3-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **When you throw a plastic bottle in the trash, where does it go? Can you trace its journey from your hand to its final destination?**

2. **What are microplastics and how do you think they end up in the ocean and eventually in the food you eat?**

3. **Only 9% of all plastic ever produced has been recycled. Does that number surprise you? Why do you think it\'s so low?**

4. **Do you think plastic pollution is a problem that individuals can solve through personal choices, or does it require systemic change? Explain your reasoning.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Plastic Production Rate
- Waste Management Capacity
- Environmental Plastic Load
- Microplastic Concentration
- Bioaccumulation in Food Webs
- Marine Ecosystem Health
- Human Health Exposure

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Business As Usual

Production Rate: +4%/year | Recycling: 9% | Policy: None

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Recycling Focus

Production Rate: +4%/year | Recycling: 27% (Tripled) | Clean-up: Active

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Production Reduction + Circular Economy

Production Rate: -40% | Recycling: 50% | Design: Circular | Policy: Producer Responsibility

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Chemical Leaching from Plastics — The release of additives like BPA, phthalates, and flame retardants from plastic as it degrades — these endocrine-disrupting chemicals enter water and food, and their concentrations in leachate increase as plastics fragment into smaller particles with greater surface area

Atmospheric Microplastic Transport — The movement of microplastic particles through the atmosphere — recent research has found microplastics in rain, snow, and air samples from remote locations, demonstrating that plastic pollution is not limited to areas near human activity but is a truly global contaminant

Extended Producer Responsibility Policy — A regulatory framework that makes manufacturers financially responsible for the entire lifecycle of their plastic products, including collection, recycling, and disposal — EPR programs in Europe have increased recycling rates to 30-40% compared to the US average of 5-6%

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Humans have produced 8.3 billion metric tons of plastic since 1950, and 6.3 billion tons of that is now waste — so where did it all go, why is it showing up inside human blood and breast milk, and can we actually solve a pollution problem this massive? Use evidence from your simulation to support your answers.

1. **Why does your model show that recycling alone cannot solve the plastic pollution crisis? What mathematical relationship between production growth and recycling capacity explains this?**

2. **How does bioaccumulation transform a seemingly low concentration of microplastics in the ocean into a significant human health concern? Trace the pathway through your model.**

3. **What does the time lag between plastic disposal and microplastic formation mean for the urgency of action? Why will microplastic levels continue rising even if we stop producing plastic today?**

4. **Compare the effectiveness of individual consumer choices versus systemic policy changes for reducing plastic pollution. What does your model suggest about where the leverage is?**

5. **How does the concept of externalized costs apply to the plastic industry — who currently pays for the environmental and health damage of plastic pollution, and who should pay?**

8 STEM Challenge: Design a Zero-Plastic-Waste Community System

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YOUR ENGINEERING MISSION

Design a comprehensive waste management and material substitution system for a community that eliminates plastic pollution pathways identified in your model, from production to environmental accumulation.

THE SCENARIO

A coastal community of 100,000 people generates 15,000 tons of plastic waste per year, of which only 8% is recycled, 15% is incinerated, and the rest goes to a landfill near the coast. Storm runoff regularly washes landfill plastic into the ocean, and local fisheries have documented increasing microplastic contamination in their catch. The community has committed to becoming zero-plastic-waste within 15 years. Your team must design a system that addresses every stage of the plastic lifecycle — from what enters the community to what leaves it — using your model\'s evidence about which intervention points are most effective.

Engineering Design Thinking

1. WHAT SPECIFIC PLASTIC ITEMS DOES YOUR PLAN ELIMINATE, SUBSTITUTE, OR REDESIGN, AND WHAT MATERIALS REPLACE THEM?

2. HOW DOES YOUR WASTE MANAGEMENT SYSTEM ACHIEVE NEAR-ZERO LEAKAGE TO THE ENVIRONMENT?

3. WHAT ECONOMIC INCENTIVES OR REGULATIONS DOES YOUR PLAN USE TO CHANGE PRODUCER AND CONSUMER BEHAVIOR?

4. HOW DOES YOUR PLAN ADDRESS THE EXISTING PLASTIC ALREADY IN THE LOCAL ENVIRONMENT AND COASTAL WATERS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

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Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental engineers who design waste management systems and develop alternatives to plastic pollution earn $65,000-$130,000/year. Materials scientists who develop biodegradable and sustainable alternatives to conventional plastics work for chemical companies, startups, and research universities, earning $70,000-$140,000/year. Environmental toxicologists who study the health effects of microplastics and persistent pollutants earn $60,000-$120,000/year at EPA, FDA, universities, and consulting firms.